Chapter 1

Matrices and Systems of Linear Equations

Probably the most important classical problem in mathematics and sciences is that of solving a system of linear equations. Many other problems can also be transferred to linear system at some stage. Linear systems arise in applications to areas such as business, economics, sociology, electronics, engineering, and physics. Therefore, it seems appropriate to begin this course with linear systems.

One most crucial tools for solving linear systems are matrices. Thus, they are firstly introduced here as follows.

1.1 Matrices

The following definitions and examples are basic but important.

Definition 1.1.1: Matrix

A *matrix of order* $m \times n$ is a collection of mn items arranged in a rectangular array with m rows and n columns.

Matrices take the following form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$$a_{ij} \longrightarrow \text{element or entry}.$$

Sometimes $A = (a_{ij})_{i,j=1}^{m,n}$, (a_{ij}) or $A_{m\times n}$ signify $m \times n$ matrix .

row = column

For $A_{m\times n}$, if $m = n$, we call A *a square matrix*.

Definition 1.1.2: Transpose of a Matrix

The *transpose* of $m \times n$ matrix A, denoted a^t, is $n \times m$ matrix formed by replacing columns of A with its rows. That is,

$$A = (a_{ij}) \quad \Longrightarrow \quad A^t = (a_{ji})$$

Example 1.1.1: Example showing transpose of a matrix

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & -1 & 0 \end{bmatrix}_{2\times 3} \quad \Longrightarrow \quad A^t = \begin{bmatrix} 1 & 3 \\ 0 & -1 \\ -2 & 0 \end{bmatrix}_{3\times 2} \quad \text{and} \quad (A^t)^t = A.$$

There are many special types of matrices that have some well-known properties. We include here the following matrices as they will be used later in this course.

Definition 1.1.3: Triangular Matrices

An $n \times n$ matrix $A = (a_{ij})$ is called *upper triangular* (สามเหลี่ยมบน) if $a_{ij} = 0$ whenever $i > j$, and *lower triangular* (สามเหลี่ยมล่าง) if ~~$a_{ij} 0$~~ ($a_{ij}=0$) whenever $i < j$. A matrix is *triangular* if it is either upper triangular or lower triangular.

Example 1.1.2: Example showing triangular matrices

An upper triangular matrix: $A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3}$

A lower triangular matrix: $A = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 2 & 3 & 0 \end{bmatrix}$

Definition 1.1.4: Symmetric Matrices

A square matrix A is called *symmetric* if $A = A^t$, that is, $a_{ij} = a_{ji}$. transpose แล้วได้เหมือนเดิม

Example 1.1.3: Example of Symmetric matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 0 & 3 \\ 1 & 3 & -1 \end{bmatrix} \text{ is symmetric.}$$

The matrix with all zero entries is called *zero matrix* or *null matrix*, denoted 0. Thus, $0 = (0)_{m\times n}$.

Definition 1.1.5: Diagonal Matrices → เมทริกซ์แนวทแยง (upper + lower)

A *diagonal matrix* has the following form:

$$A_n = \begin{bmatrix} a_{11} & & & 0 \\ & a_{22} & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{bmatrix}$$

An *identity matrix* is a special diagonal matrix of the form

$$I_n = \begin{bmatrix} 1 & & & 0 \\ & 1 & & \\ & & \ddots & \\ 0 & & & 1 \end{bmatrix}$$

Example 1.1.4: Example of identity matrices → เมทริกซ์เอกลักษณ์

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

One special type of matrices whose number of column or row is one is vector. This notion is used very commonly in mathematics.

Definition 1.1.6: Vector

An $m \times 1$ matrix is called a *column vector*: $\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}$.

A $1 \times m$ matrix is called a *row vector*: $\vec{b}^t = \begin{bmatrix} b_1 & b_2 & \cdots & b_m \end{bmatrix}$.

As it will be used later, we also introduce here unit vectors.

Definition 1.1.7: Unit Vectors ⟶ เวกเตอร์ 1 หน่วย

The column vectors containing 1 in the i-th position and 0 elsewhere are called *standard unit vectors* or the canonical vectors in $\mathbb{R}^n$:

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \cdots, \quad \vec{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

1.2 Matrix Addition and Scalar Multiplication

In this section, we define arithmetic operations with matrices and look at some of their algebraic properties. As mentioned, matrices are one of the most powerful tools in mathematics. To use them effectively, we must be adept at matrix arithmetic.

Definition 1.2.1: Addition and Subtraction of Matrices ⟶ การบวก-ลบ เมทริกซ์

$A = (a_{ij})$ and $B = (b_{ij})$ are $m \times n$ matrices.

- The *sum* of A and B is the $m \times n$ matrix $C = A + B$, whose ij-th entry is $c_{ij} = a_{ij} + b_{ij}$.
- The *difference* of A and B, $C = A - B$, has ij-th entry $c_{ij} = a_{ij} - b_{ij}$.

Definition 1.2.2: Scalar multiplication

$A = (a_{ij})$ a matrix and α a real number (scalar).

$$C = \alpha A \quad \text{has } ij\text{-th entry} \quad c_{ij} = \alpha a_{ij}.$$

Example 1.2.1: Example of Addition, subtraction and scalar Multiplication

$$\text{Consider} \quad A = \begin{bmatrix} 0 & -2 \\ 1 & 4 \\ 3 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 7 \\ 1 & -3 \\ 2 & -1 \end{bmatrix}. \quad \text{Then}$$

$$A + B = \begin{bmatrix} 0 & 5 \\ 2 & 1 \\ 5 & 1 \end{bmatrix}, \quad A - B = \begin{bmatrix} 0 & -9 \\ 0 & 7 \\ 1 & 3 \end{bmatrix}, \quad 2A = \begin{bmatrix} 0 & -4 \\ 2 & 8 \\ 6 & 4 \end{bmatrix}.$$

Theorem 1.2.1: Basic operation with matrices

For matrices A, B and C with dimention $m \times n$ and α, β two real numbers, the following hold:

กฎการเปลี่ยนกลุ่ม

1. (Associativity): $A + (B + C) = (A + B) + C$.

กฎการสลับที่การบวก

2. (Commutativity): $A + B = B + A$.

การมีอยู่ของเอกลักษณ์การบวก

3. (Additive Identity): $A + 0 = 0 + A$.

การมีอยู่ของ inverse การบวก

4. (Additive Inverse): $A + (-A) = 0 = (-A) + A$.

กฎการแจกแจง

5. (Distributive laws): $(\alpha + \beta)A = \alpha A + \beta A$ & $\alpha(A + B) = \alpha A + \alpha B$.

สมบัติ transpose ของเมทริกซ์

6. $(\alpha A + \beta B)^t = \alpha A^t + \beta B^t$. → $(2A - 3B)^t = (2A)^t + (-3B)^t$
$= 2(A^t) + (-3)(B^t)$
$= 2A^t - 3B^t$

1.3 Matrix Multiplication

The multiplication of matrices are unlike the usual multiplication of numbers. In fact, it is possible to multiply a matrix A times a matrix B is the number of columns of A equals the number of rows of B. The first column of the product is determined by the

first column of B, the second column by the the second column of B, and so on. Thus, to determine the (i, j) entry of the product AB, we use the entries of the ith row of A and the jth column of B.

Example 1.3.1: Example of matrix multiplication

Consider $A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}_{2\times 2}$ and $B = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 4 & 5 \end{bmatrix}_{2\times 3}$. Then

$$AB = \begin{bmatrix} -1 & 4 & 5 \\ -1 & 12 & 17 \end{bmatrix} \quad \text{But} \quad BA = \text{Invalid}.$$

Definition 1.3.1: Multiplication of matrices

Given A an $m \times n$ matrix and B an $n \times p$ matrix. The product $AB := C$ is an $m \times p$ matrix $\longrightarrow$ $C = (c_{ij})$ with entries defined below.

$$c_{ij} = \boxed{a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Note that $AA = A^2$.

What this definition says is that to find ijth element of the product, we take the ith row of A and the jth column of B, multiply the corresponding elements pairwise, and add the resulting numbers.

In order to pair off the elements in this way, the number columns of A must be the same as the number of rows of B. If this does not satisfy, the multiplication is invalid.

Example 1.3.2

Given $A = \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ -2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 1 \\ 0 & -3 \end{bmatrix}$. Then

BA is Invalid. However, $AB = \begin{bmatrix} 5 & 4 \\ 15 & 3 \\ -10 & -14 \end{bmatrix}_{3\times 2}$.

Remark: In general, AB and BA may not be equal, even when both are valid.

Theorem 1.3.1: Basic operation with matrices

Let A, B and C be matrices such that all product and sums below are defined and k be a number. Then

1. $A(BC) = (AB)C$.
2. (Commutativity): $(A + B)C = AC + BC$.
3. $C(A + B) = CA + CB$
4. $k(AB) = (kA)B = A(kB)$.

1.4 System of Linear Equations

In this section, we introduce system of linear equations. We begin with the definition of a linear equation.

A *linear equation* in variables $x_1, \ldots, x_n$ is

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where *constant term* b and *coefficients* $a_1, \ldots, a_n$ are real numbers.

Example 1.4.1

Determine whether the following equations are linear.

1. $2x_1 - 6x_2 + 7 = x_3 \longrightarrow 2x_1 - 6x_2 - x_3 = -7$ (linear)
2. $\pi x_2 - \sqrt{5}x_1 = x_3 \longrightarrow -\sqrt{5}x_1 + \pi x_2 - x_3 = 0$ (linear)
3. $4x_1x_2 = x_3 \longrightarrow$ (non-linear)
4. $x_1 = \sqrt{x_2} + 4 \longrightarrow$ (non-linear)

When we consider a multiple linear equations to find common solutions for them, it means we are considering a linear system.

A system of *linear* equations

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}\right\} \tag{1.1}$$

$A\vec{x} = \vec{b}$
$A^{-1}(A\vec{x}) = A^{-1}\vec{b}$
$(A^{-1}A)\vec{x} = A^{-1}\vec{b}$
$I\vec{x} = A^{-1}\vec{b}$
$\vec{x} = A^{-1}\vec{b}$

with n variables and m equations.

If $m = n$, *square* system.

If $m \neq n$, *rectangular* system.

The (1.1) can be written as $Ax = b$, where $A = (a_{ij})$ is the $m \times n$ matrix of the coefficient a_{ij}, $x = (x_1, x_2, \ldots, x_n)^t$ is the $n \times 1$column vector of the variables and $b = (b_1, b_2, \ldots, b_m)^t$ is the $m \times 1$ column vector of constants.

Example 1.4.2

$$\left.\begin{aligned} 3x + y - z &= 5 \\ 2y + 4z &= 0 \end{aligned}\right\} \quad \text{equivalent to} \quad \begin{bmatrix} 3 & 1 & -1 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$\left.\begin{aligned} x + 5y &= 1 \\ -x + y - z &= -2 \\ -y + 3z &= 10 \end{aligned}\right\} \quad \text{equivalent to} \quad \begin{bmatrix} 1 & 5 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 10 \end{bmatrix}$$

For any linear system, there are three possibilities of the solutions.

Three possibilities of the set of solutions

1. *Unique Solution*: Only one vector $\vec{x}$ such that $A\vec{x} = \vec{b}$
2. *No Solution*: No vector $\vec{x}$ such that $A\vec{x} = b\vec{b}$
3. *Infinitely many solution*: If a system $A\vec{x} = \vec{b}$ has two solutions, u_1 & u_2, then every combination $\beta u_1 + (1-\beta)u_2$, $\beta \in [0,1]$ is also a solution.

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Example 1.4.3

Find solution(s) of the following linear systems.

1. $4x + 3y = 11 \quad (1)$

   $4x - 3y = 5 \quad (2)$

   $(1) + (2); \quad 8x = 16 \implies x = 2$

   $(2); \quad 8 - 3y = 5 \implies y = 1$

   Unique Solution

2. $4x + 3y = 11 \quad (1)$

   $8x + 6y = 22 \quad (2)$

   $2 \times (1); \quad 8x + 6y = 22 \quad$ (this is (2))

   Infinitely many Solution.

   Consider $4x + 3y = 11.$

   Let $x = \alpha \in \mathbb{R}$, then $y = \dfrac{11 - 4\alpha}{3}$.

   Therefore, vector $\left(\alpha, \dfrac{11 - 4\alpha}{3}\right)$ is solution.

3. $4x + 3y = 11 \quad (1)$

   $8x + 6y = 20 \quad (2)$

   $2 \times (1); \quad 8x + 6y = 22 \quad (3)$

   $(3) - (2); \quad 0 = 2 \quad$ *Impossible*

   No Solution

Definition 1.4.1

A linear system $Ax = b$ is said to be *consistent* if it has at least one solution and is said to be *inconsistent* otherwise.

Recall (1.1) which can be written as $A\vec{x} = \vec{b}$.

Augmented matrix associated with (1.1) is

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] = \left[\begin{array}{c|c} A & \vec{b} \end{array}\right]$$

1.5 Gaussian Elimination

Any linear system can be solved by performing operations on the augmented matrix. To obtain an easier equivalent system, the following row operations may be applied to the augmented matrix.

$$A\vec{x} = \vec{b} \quad \sim \quad \text{An equivalent system that is easier to solve.}$$

Definition 1.5.1: An elementary row operation

1. *Interchanging*: interchange two rows of the matrix, say row p and row q, denoted by R_{pq}.
2. *Scaling*: Multiply row p by a nonzero number c, denoted by cR_p.
3. *Replacement*: replace a row p by the sum of that row and a scalar multiple c of another row q, denoted by $R_p + cR_q$.

Matrices A and B are *row equivalent* if one can be changed to the other by a sequence of elementary row operations $A \sim B$.

Example 1.5.1

$$\text{Let } A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 4 & 5 \\ -2 & 1 & 0 & 3 \end{bmatrix}. \qquad \text{Then } A \sim \begin{bmatrix} 0 & 1 & 4 & 5 \\ 1 & 0 & -1 & 2 \\ -2 & 1 & 0 & 3 \end{bmatrix} R_{12}$$

$$A \sim \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 4 & 5 \\ -1 & \frac{1}{2} & 0 & \frac{3}{2} \end{bmatrix} \frac{1}{2}R_3 \qquad A \sim \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 4 & 5 \\ 0 & 1 & -2 & 7 \end{bmatrix} R_3 + 2R_1$$

Observation: The solution of the system $A\vec{x} = \vec{b}$ remains unchanged as long as we apply the elementary row operations to $[A|\vec{b}]$.

Example 1.5.2

Consider $\left.\begin{array}{rl} x_1 + x_2 - x_3 & = 1 \\ 2x_1 + x_2 - 3x_3 & = 2 \\ x_2 - 5x_3 & = 6 \end{array}\right\}$. Find solution(s) for this system.

Solution

$$\left[\begin{array}{c|c} A & \vec{b} \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & 1 & -3 & 2 \\ 0 & 1 & -5 & 6 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 1 & -5 & 6 \end{array}\right] R_2 + (-2R_1)$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & -6 & 6 \end{array}\right] \begin{array}{l} R_1 + R_2 \\ \\ R_3 + R_2 \end{array} \sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{array}\right] \begin{array}{r} \\ -R_2 \\ -\frac{1}{6}R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right] \begin{array}{r} R_1 + 2R_3 \\ R_2 + (-R_3) \\ \\ \end{array}$$

This is the augmented matrix of

$$\begin{array}{rcl} x_1 & = & -1 \\ x_2 & = & 1 \\ x_3 & = & -1 \end{array}.$$

Therefore, the solution of the system is $\{(-1, 1, -1)\}$.

Example 1.5.3

$$\begin{aligned} x_1 - x_2 &= 0 \\ -2x_2 + 4x_3 &= 3 \\ x_1 + x_2 - 4x_3 &= 1 \end{aligned}$$

Solution

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & -2 & 4 & 3 \\ 1 & 1 & -4 & 1 \end{array}\right] \begin{array}{c} R_3 + (-R_1) \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & -2 & 4 & 3 \\ 0 & 2 & -4 & 1 \end{array}\right]$$

This is the augmented matrix associated with

$$\begin{aligned} x_1 - x_2 &= 0 \quad (1) \\ -2x_2 + 4x_3 &= 3 \quad (2) \\ 2x_2 - 4x_3 &= 1 \quad (3) \end{aligned}$$

(2) + (3); $0 = 4$ Impossible. No Solution

At a certain step, we may encounter with a *triangular system* which is the form in the kth equation the coefficients of the first $k-1$ variables are all zero and the coefficient of x_k is nonzero.

To solve such the form, we refer to the method of solving as *back substitution*.

$$\left[\begin{array}{c|c} A & b \end{array}\right] \sim \left[\begin{array}{cccc|c} u_{11} & u_{12} & \cdots & u_{1n} & b_1 \\ 0 & u_{22} & \cdots & u-2n & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & u_{nn} & b_n \end{array}\right] \text{where} u_{ii} \neq 0.$$

Then apply back substitution to obtain the solution

Step 1: $x_n = \dfrac{b_n}{u_{nn}}$

Step 2: Recursively compute

$$x_i = \frac{1}{u_{ii}}\left(b_i \sum_{j=i+1}^{n} u_{ij}x_j\right), \quad \text{for } i = n, n-1, \cdots, 2, 1.$$

Example 1.5.4

Consider the system $A\vec{x} = \vec{b}$, where $A = \begin{bmatrix} 2 & -2 & -5 & -3 \\ 2 & -1 & -3 & 2 \\ 4 & -1 & -4 & 10 \\ 0 & 1 & 2 & 5 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 9 & 4 & 4 & -5 \end{bmatrix}$. Then

$$\left[A|\vec{b}\right] = \left[\begin{array}{cccc|c} 2 & -2 & -5 & -3 & 9 \\ 2 & -1 & -3 & 2 & 4 \\ 4 & -1 & -4 & 10 & 4 \\ 0 & 1 & 2 & 5 & -5 \end{array}\right] \sim \left[\begin{array}{cccc|c} 2 & -2 & -5 & -3 & 9 \\ 0 & 1 & 2 & 5 & -5 \\ 0 & 3 & 6 & 16 & -14 \\ 0 & 1 & 2 & 5 & -5 \end{array}\right] \begin{array}{l} \\ R_2 + (-R_1) \\ R_3 + (-2R_1) \\ \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 2 & -2 & -5 & -3 & 9 \\ 0 & 1 & 2 & 5 & -5 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ \\ R_3 + (-3R_2) \\ R_4 + (-R_2) \end{array}$$

This implies that

R_4; $x_4 = 1$

R_2; $x_2 + 2x_3 + 5x_4 = -5 \implies x_2 + 2x_3 = -10.$

Set $x_3 = \alpha$ as a free variable. Then $x_2 = -10 - 2\alpha$

R_1; $2x_1 - 2x_2 - 5x_3 - 3x_4 = 9 \implies 2x_1 - 2(-10 - 2) - 5\alpha - 3 = 9$

$$2x_1 + 20 + 4\alpha - 5\alpha = 12 \implies x_1 = -4 + \frac{\alpha}{2}$$

The solution is $\left\{\left(-4 + \frac{\alpha}{2}, -10 - 2\alpha, \alpha, 1\right)\right\}$

$$\vec{v} = \begin{bmatrix} -4 \\ -10 \\ 0 \\ 1 \end{bmatrix} + \alpha \begin{bmatrix} \frac{1}{2} \\ -2 \\ 1 \\ 0 \end{bmatrix}, \quad \alpha \in \mathbb{R}. \text{ Infinitely many solution.}$$

The following form can also considered as an important tool for solving a linear system.

Definition 1.5.2: Row Echelon Form

An $m \times n$ matrix U is said to be in *row echelon form* if it satisfies:

1. All zero rows must be below non-zero rows.
2. Each non-zero row has the non-zero *leading entry*. It must be on the right hand side of the leading entry of the previous row.
3. All entries below the leading entry in each column must be zero.

This process is called **Gaussian elimination**.

Example 1.5.5

$$A = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad \text{is in row echelon form.}$$

$$B = \begin{bmatrix} 3 & 0 & 1 & 4 \\ 5 & 1 & 0 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix} \quad \text{is not (yet) in row echelon form.}$$

Definition 1.5.3: Reduced Row Echelon

A row echelon form matrix is said to be in *reduced row echelon form* if

1. All leading entry must be 1.
2. For the column containing the leading entry, all other numbers must be 0.

This process is called **Gauss-Jordan reduction**.

Example 1.5.6

$$A = \begin{bmatrix} 1 & 0 & 4 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{is in reduced row echelon form.}$$

$$B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{echelon form but not reduced.}$$

Remark: Every Matrix can be turned into (reduced) row echelon form by elementary row operations.

Remark: We can also call the leading entry a *pivot*.

Example 1.5.7

Turn matrix A to be in reduced row echelon form

$$A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix} R_{14}$$

$$\sim \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 5 & 10 & -15 & -15 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix} \begin{matrix} \\ R_2 + R_1 \\ R_3 + 2R_1 \\ \\ \end{matrix} \sim \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix} \begin{matrix} \\ \frac{1}{2}R_2 \\ \frac{1}{5}R_3 \\ \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 & 0 \end{bmatrix} \begin{matrix} \\ \\ R_3 - R_2 \\ R_4 + 3R_2 \end{matrix} \sim \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} R_{34}$$

$$\sim \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} -\frac{1}{5}R_3 \sim \begin{bmatrix} 1 & 0 & -3 & 3 & 5 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} -R_1 - 4R_2$$

$$\sim \begin{bmatrix} 1 & 0 & -3 & 0 & 5 \\ 0 & 1 & 2 & 0 & -3 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 - 3R_3 \\ R_2 + 3R_3) \\ \\ \\ \end{matrix}$$

Reduced echelon form

Consider

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \sim \begin{array}{rcrcl} x_1 & & & 2x_3 & = 2 \\ & x_2 & + & x_3 & = 1 \\ & & & 0 & = 0 \end{array}$$

$$x_1 = 2 + 2x_3$$
$$x_2 = 1 - x_3$$

The solution of this system is in the form

$$\begin{array}{c} x_1 = 2 + 2t, \quad t \in \mathbb{R} \\ x_2 = 1 - t, \\ x_3 = t \end{array} \sim \begin{array}{c} x_1 = 2 + 2x_3, \qquad t \in \mathbb{R} \\ x_2 = 1 - x_3, \\ x_3 \text{ is a free variable} \end{array}$$

is called “the general solution" (infinitely many solution)

Remark:[Existance Theorem] The system of linear equations has a solution if and only if the echelon form of the augmented matrix *does not* have a row $\left[\begin{array}{ccc|c} 0 & 0 & \cdots & 0 \end{array} \middle| \begin{array}{c} b \end{array}\right]$, $b \neq 0$.

Summary

To find solutions set of a linear system, we may proceed by the following steps.

1. Write the augmented matrix.
2. Apply elementary row operations until the matrix turned into row echelon form.
3. If we know that the system has a solution, we will try to turn the matrix into the reduced row echelon form. Moreover, we call the variables correspond to the pivots “basic variables" and the other variables “free variables".
4. Write the general solution.

Example 1.5.8

Solve the following systems.

$$\begin{cases} x_1 + 2x_2 + x_3 + 3x_4 = 12 \\ \quad 3x_2 + x_3 + 2x_4 = 7 \\ 2x_1 + x_2 + x_3 + 4x_4 = 21 \end{cases}$$

Solution:

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 12 \\ 0 & 3 & 1 & 2 & 7 \\ 2 & 1 & 1 & 4 & 21 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 12 \\ 0 & 3 & 1 & 2 & 7 \\ 0 & -3 & -1 & -2 & -3 \end{array}\right] \begin{array}{l} \\ \\ R_3 + (-2R_1) \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 12 \\ 0 & 3 & 1 & 2 & 7 \\ 0 & 0 & 0 & 0 & 4 \end{array}\right] \begin{array}{l} \\ \\ R_3 + R_2 \end{array}$$

Therefore, no solution.

Example 1.5.9

Solve the following systems.

$$\begin{cases} x_1 - 2x_2 + 2x_3 - x_4 = -3 \\ 2x_1 - 4x_2 + 2x_3 + 2x_4 = 8 \\ 2x_1 - 4X_2 + 3X_3 \quad\quad = 1 \end{cases}$$

Solution:

$$\left[\begin{array}{cccc|c} 1 & -2 & 2 & -1 & -3 \\ 2 & -4 & 2 & 2 & 8 \\ 2 & -4 & 3 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & -2 & 2 & -1 & -3 \\ 0 & 0 & -2 & 4 & 14 \\ 0 & 0 & 1 & -2 & -7 \end{array}\right] \begin{array}{l} \\ R_2 + (-2R_1) \\ R_3 + (-R_2) \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -2 & 2 & -1 & -3 \\ 0 & 0 & 1 & -2 & -7 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ -\frac{1}{2}R_2 \\ R_3 + \frac{1}{2}R_2 \end{array} \sim \left[\begin{array}{cccc|c} 1 & -2 & 0 & 3 & 11 \\ 0 & 0 & 1 & -2 & -7 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} R_1 + (-2R_2) \end{array}$$

Therefore,

$$\begin{array}{rcrcrcl} x_1 & - & 2x_2 & & + & 3x_4 & = & 11 \\ & & & x_3 & - & 2x_4 & = & -7 \end{array} \quad \Longrightarrow \quad \begin{array}{c} x_1 = 11 + 2x_2 - 3x_4 \\ x_3 = -7 + 2x_4 \\ x_2 \text{ \& } x_4 \text{are free variables.} \end{array}$$

Example 1.5.10

Consider the following

$$\left[\begin{array}{cccc|c} 2 & -4 & 2 & -2 & \\ 2 & -4 & 3 & -4 & \\ 4 & -8 & 3 & -2 & \\ 0 & 0 & -1 & 2 & \end{array}\right] \sim \left[\begin{array}{cccc|c} 2 & -4 & 2 & -2 & \\ 0 & 0 & 1 & -2 & R_3 + (-R_1) \\ 0 & 0 & -1 & 2 & R_3 + (-2R_1) \\ 0 & 0 & -1 & 2 & \end{array}\right]$$

$$\sim \left[\begin{array}{cccc|c} 2 & -4 & 2 & -2 & \\ 0 & 0 & 1 & -2 & R_3 + R_2 \\ 0 & 0 & 0 & 0 & R_4 + R_2 \\ 0 & 0 & 0 & 0 & \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & -2 & 1 & -1 & \frac{1}{2}R_1 \\ 0 & 0 & 1 & -2 & \\ 0 & 0 & 0 & 0 & \\ 0 & 0 & 0 & 0 & \end{array}\right]$$

$$\sim \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & R_1 + (-R_2) \\ 0 & 0 & 1 & -2 & \\ 0 & 0 & 0 & 0 & \\ 0 & 0 & 0 & 0 & \end{array}\right]$$

Therefore,

$$\begin{array}{l} x_1 - 2x_2 \quad\quad = 1 \\ \qquad\quad x_3 = -2 \end{array} \implies \begin{array}{c} x_1 = 11 + 2x_2 \\ x_3 = -2 \\ x_2 \text{ is a free variable.} \end{array}$$

Example 1.5.11

Find the value of h for which $\left[\begin{array}{cc|c} 1 & -3 & h \\ -2 & 6 & -5 \end{array}\right]$ is the augmented matrix of a consistent linear system

Solution:

$$\left[\begin{array}{cc|c} 1 & -3 & h \\ 0 & 0 & 2h-5 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -3 & h \\ 0 & 0 & 2h-5 \end{array}\right] \quad R_2 + 2R_1$$

$$\implies \quad 2h - 5 = 0 \quad \implies \quad h = \frac{5}{2}.$$

1.6 Inverse of a Matrix

This section discusses about inverse of a matrix. It is crucial and useful in some certain occasion involving matrices due to its properities.

Definition 1.6.1

The $n \times n$ matrix $I_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$ is the *identity matrix* of order n.

Definition 1.6.2

Let A be an $n \times n$ matrix. If we have an $n \times n$ matrix B such that $AB = BA = I_n$, then we call B the *inverse* of A and we say that A is *invertible*. Denote B by A^{-1}.

Theorem 1.6.1

If a matrix A is invertible, then the equation $A\vec{x} = \vec{b}$ has exactly one solution, that is $\vec{x} = A^{-1}\vec{b}$.

We note that to find A^{-1} by using row operations

$$\left[A \middle| I_n\right] \quad \sim \quad \left[I_n \middle| A^{-1}\right]$$

Example 1.6.1

Find A^{-1} where $A = \begin{bmatrix} -1 & -1 & 2 \\ 2 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$

Solution:

$$\left[\begin{array}{ccc|ccc} -1 & -1 & 2 & 1 & 0 & 0 \\ 2 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & -1 & 0 & 0 \\ 0 & -1 & 2 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right] \begin{matrix} -R_1 \\ R_2 + 2R_1 \\ R_3 + R_1 \end{matrix}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & -2 & -2 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right] \begin{matrix} R_1 + R_2 \\ -R_2 \end{matrix}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right] R_2 + 2R_3$$

Therefore, $A^{-1} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$.

Theorem 1.6.2

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ Then A is invertible if and only if $ad - bc \neq 0$, and the inverse

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Note: An invertible matrix is also called *non-singular*.

1.7 Determinant

With each square matrix it is possible to associate a real number called the determinant of matrix. The value of this number will tell us whether or not the matrix is singular.

Given $A = (a_{ij})_{n \times n}$, let $M_{ij}(A)$ be the matrix obtained by deleting the row i^{th} and the column j^{th} of A.

Definition 1.7.1: The determinant of A (det A or $|A|$)

If $n = 1$, $A = [a_{11}]$ then $\det A = a_{11}$. If $n > 1$, $\det A = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det M_{1j}(A)$.

Remark: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \Longrightarrow \quad \det A = ad - bc.$

Example 1.7.1

Given $A = \begin{bmatrix} 1 & 5 & 0 \\ 0 & -3 & 1 \\ 2 & 4 & -1 \end{bmatrix}$

Solution:

$$\det A = 1(-1)^{1+1} \begin{vmatrix} -3 & 1 \\ 4 & -1 \end{vmatrix} + 5(-1)^{1+2} \begin{vmatrix} 0 & 1 \\ 2 & -1 \end{vmatrix} + 0(-1)^{1+3} \begin{vmatrix} 0 & -3 \\ 2 & 4 \end{vmatrix}$$
$$= (-1) + 10 + 0 = 9.$$

Definition 1.7.2: The (i, j)-cofactor of A

$C_{ij}(A) = (-1)^{1+j} \det M_{1j}(A)$ is the (i, j)-cofactor of A.

Therefore, $\det A = \sum_{j=1}^{n} a_{ij} C_{ij}(A)$

Remark: Cofactor expansion across any row of A works as well.

Theorem 1.7.1

$$\det A = \sum_{j=1}^{n} a_{ij} C_{ij}(A),$$

for all $i = 1, 2, \cdots, n$.

Note that one can often save work by expanding along the row or column that contains the most zeros.

Example 1.7.2

Let $\begin{bmatrix} 1 & 5 & 8 & 9 & 0 \\ 0 & -3 & -5 & 7 & 1 \\ 0 & 0 & 1 & 5 & 0 \\ 2 & 4 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$. Find $\det A$.

Solution:

$$\det A = (-2)C_{54}(A) = (-2)(-1)^{5+4}\begin{vmatrix} 1 & 5 & 8 & 0 \\ 0 & -3 & -5 & 1 \\ 0 & 0 & 1 & 0 \\ 2 & 4 & 2 & -1 \end{vmatrix}$$

$$= (-2)(-1)(1)(-1)^{3+3}\begin{vmatrix} 1 & 5 & 0 \\ 0 & -3 & 1 \\ 2 & 4 & -1 \end{vmatrix} = 2(9) = 18.$$

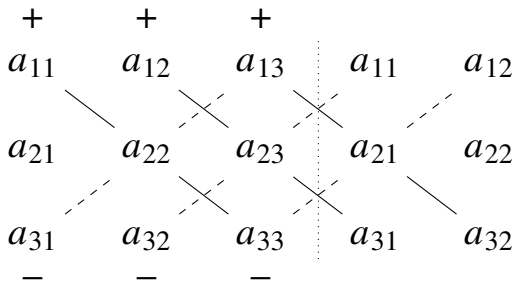
Corollary 1.7.2

If A is a triangular matrix, then $\det A$ is the product of all diagonal entries.

Example 1.7.3

Given $A = \begin{bmatrix} -4 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 5 & -1 \end{bmatrix}$. Then $\det A = (-4) \times (2) \times (-1) = 8$.

For a 3×3 matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, the determinant of A can be computed as follows:



$$\begin{aligned}\det A = &(a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}) \\ &- (a_{31}a_{22}a_{13} + a_{32}a_{23}a_{11} + a_{33}a_{21}a_{12}) .\end{aligned}$$

The following theorems are for important properties of determinant of matrices.

Theorem 1.7.3

Let A be a square matrix.

1. $\det A = \det A^t$.
2. If C is obtained from A by the row operation R_{pq} $(p \neq q)$, then $\det C = -\det A$.
3. If A contains two identical rows, then $\det A = 0$.
4. If C is obtained from A by the raw operation $R_p + \alpha R_q$ $(p \neq q)$, then $\det C = \det A$.
5. If C is obtained from A by the raw operation αR_p, then $\det C = \alpha \det A$.

Theorem 1.7.4

Let A be a square matrix.

1. A is invertible if and only if $\det A \neq 0$.
2. If $\det A \neq 0$, then $\det A^{-1} = \dfrac{1}{\det A}$.

Theorem 1.7.5: L

t A, B be $n \times n$ matrices. Then

$$\det (AB) = (\det A) (\det B) .$$

Corollary 1.7.6

Let A, B be $n \times n$ matrices. Then

1. $\det\left(A^k\right) = (\det A)^k$.
2. $\det(\alpha A) = \alpha^n \det(A)$.

Example 1.7.4

Given $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 5.$

Find 1. $\begin{vmatrix} 2a & 2b & 2c \\ g & h & i \\ d & e & f \end{vmatrix}$ 2. $\begin{vmatrix} a & b & c \\ 2a+d & 2b+e & 2c+f \\ 3g & 3h & 3i \end{vmatrix}$.

1. $\begin{vmatrix} 2a & 2b & 2c \\ g & h & i \\ d & e & f \end{vmatrix} \overset{2R_1}{=} \begin{vmatrix} a & b & c \\ g & h & i \\ d & e & f \end{vmatrix} \overset{R_{2,3}}{=} 2(-1)\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -10.$

2. $\begin{vmatrix} a & b & c \\ 2a+d & 2b+e & 2c+f \\ 3g & 3h & 3i \end{vmatrix} \overset{R_2+2R_1}{=} \begin{vmatrix} a & b & c \\ d & e & f \\ 3g & 3h & 3i \end{vmatrix} \overset{3R_3}{=} 3\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$

$= 3(5) = 15.$

Example 1.7.5

Let A, B, C be 3×3 matrices with $\det(A) = 2$, $\det(B) = 3$ and $\det(C) = 4$. Find $\det\left(2AB^tC^{-1}\right)$.

Solution:

$$\det\left(2AB^tC^{-1}\right) = 2^3\det(A)\left(\det B^t\right)\left(\det C^{-1}\right) = 8\det(A)(\det B)\frac{1}{\det C} = \frac{8 \cdot 2 \cdot 3}{4} = 12.$$

We define a new matrix called the *adjoint* of A.

Definition 1.7.3: Adjoint of a Matrix

For a matrix $A = \left(a_{ij}\right)_{n\times n}$ and $C_{ij}(A)$ is the (i, j)-cofactor of A, *adjoint matrix* of A is

$$\text{adj}(A) = \left[C_{ij}(A)\right]^t$$

The adjoint matrix has an important property as follows.

Theorem 1.7.7

If A is an $n \times n$ matrix, then

$$A\,(\text{adj}(A)) = (\text{adj}(A))\,A = \det(A) I_n.$$

From this theorem, we have a formula for A^{-1}.

Corollary 1.7.8

If A is an $n \times n$ matrix with $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det(A)}\text{adj}(A).$$

1.8 Cramer's Rule

Using the formula in the previous Corollary, we can derive a rule for representing the solution to the system $A\vec{x} = \vec{b}$ in terms of determinants.

Let A be an $n \times n$ matrix with $\det A \neq 0$, and $\vec{b}$ a vector in $\mathbb{R}^n$. Then the solution for $A\vec{x} = \vec{b}$ is

$$x_i = \frac{\det A_i}{\det A},$$

where A_i is obtained from A by replacing the ith column of A by b.

Example 1.8.1

Use Cramer's Rule to solve
$$\begin{aligned} x_1 - x_2 + 2x_3 &= -2 \\ 3x_1 - 2x_2 + 4x_3 &= -5 \\ 2x_2 - 5x_3 &= 2. \end{aligned}$$

Solution:

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & -2 & 4 \\ 0 & 2 & -5 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} -2 \\ -5 \\ 2 \end{bmatrix},$$

$$A_1 = \begin{bmatrix} -2 & -1 & 2 \\ -5 & -2 & 4 \\ 2 & 2 & -5 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & -2 & 2 \\ 3 & -5 & 4 \\ 0 & 2 & -5 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & -1 & -2 \\ 3 & -2 & -5 \\ 0 & 2 & 2 \end{bmatrix}$$

$$\det A = 1, \quad \det A_1 = 1, \quad \det A_2 = -1 \quad \text{and} \quad \det A_3 = 0.$$

Therefore,

$$x_1 = \frac{\det A_1}{\det A} = -1, \quad x_2 = \frac{\det A_2}{\det A} = 1, \quad x_3 = \frac{\det A_3}{\det A} = 0.$$

The solution is $\vec{x} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$.

1.9 Homogeneous Linear System

Consider the system of linear equation

$$A\vec{x} = \vec{0}_m$$

where A is an $m \times n$**matrix**, $\quad \vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \vec{0}_m = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \longrightarrow m$ entries.

There is always at least one solution for this system, where is $\vec{0}_n$, called a *trivial solution*. Other kinds of solution are called *non-trivial*.

The general solution of a homogeneous linear system can be written in the form

$$\vec{x} = t_1\vec{v}_1 + t_2\vec{v}_2 + \cdots + T_p\vec{v}_p,$$

where t_i is a free variable (parametric vector form).

Example 1.9.1

Solve
$$\begin{aligned} x_1 - 3x_2 - 2x_3 &= 0 \\ x_2 - x_3 &= 0 \\ -2x_1 + 3x_2 + 7x_3 &= 0 \end{aligned}$$

Solution:

$$\begin{bmatrix} 1 & -3 & -2 \\ 0 & 1 & -1 \\ -2 & 3 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -2 \\ 0 & 1 & -1 \\ 0 & -3 & 3 \end{bmatrix} \underset{R_3 + 2R_1}{\sim} \begin{bmatrix} 1 & -3 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \underset{R_3 + 3R_2}{} \sim \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \underset{R_1 + 3R_1}{}$$

Thus,

$$\begin{aligned} x_1 \quad -5x_3 &= 0 \\ x_2 - x_3 &= 0 \end{aligned} \implies \begin{aligned} x_1 &= 5x_3 \\ x_2 &= x_3 \\ x_3 &\text{ is a free variable.} \end{aligned}$$

$$\text{Therefore } \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5x_3 \\ x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}.$$

$$\text{Therefore} \quad \vec{x} = t \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

From the above discussion, we record the result as follows.

Theorem 1.9.1

If a non-homogenous system $A\vec{x} = \vec{b}$ has a solution $\vec{p}$, namely particular solution, then the general solution for $A\vec{x} = \vec{b}$ is in the form $\vec{x} = \vec{p} + \vec{v}_h$, where $\vec{v}_h$ is the general solution for the corresponding to homogenous system $A\vec{x} = \vec{0}$.

1.10 Exercise 1

1. Find w, x, y and z in the following:

$$2\begin{bmatrix} x+5 & y-3 \\ 5 & 7+2z \end{bmatrix} = \begin{bmatrix} 2 & 3-y \\ w+7 & 14+z \end{bmatrix}$$

2. Using elementary row operations to solve the linear equation

$$\begin{aligned} 2x_1 - x_2 &= 0, \\ -x_1 + 2x_2 + x_3 &= 6, \\ -x_1 - x_3 &= -4. \end{aligned}$$

3. Find an echelon form of $Ax = b$ and solve the system:

$$A = \begin{bmatrix} 2 & 6 & 1 \\ 3 & 9 & 2 \\ 0 & -1 & 3 \end{bmatrix} \text{ and } b = \begin{bmatrix} 5 \\ 8 \\ 3 \end{bmatrix}$$

4. Find an echelon form of $Ax = b$ and solve the system:

$$A = \begin{bmatrix} 0 & 2 & 5 \\ -1 & 7 & -5 \\ -1 & 8 & 3 \end{bmatrix} \text{ and } b = \begin{bmatrix} 15 \\ -20 \\ 4 \end{bmatrix}$$

5. Find the sequences of elementary operations that will turn A to a reduce echelon matrix, where A is the following:

$$A = \begin{bmatrix} 3 & 0 & 2 & -1 \\ 2 & 3 & -1 & 0 \\ 1 & 5 & 0 & 8 \end{bmatrix}$$

6. Verify whether $Ax = b$ is consistent for the following A and b and find a solution if it exists:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -4 \\ 0 & 1 & -3 \\ -1 & 0 & -2 \end{bmatrix} \text{ and } b = \begin{bmatrix} 1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

7. Find the values of a for which the system

$$\begin{aligned} x_1 + x_2 - x_3 &= 3 \\ x_1 - x_2 + 3x_3 &= 4 \\ x_1 + x_2 + \left(a^2 - 10\right) x_3 &= a \end{aligned}$$

(1) has no solution (2) has the unique solution (3) has infinitely many solutions.

1. Find w, x, y and z in the following:

$$2\begin{bmatrix} x+5 & y-3 \\ 5 & 7+2z \end{bmatrix} = \begin{bmatrix} 2 & 3-y \\ w+7 & 14+z \end{bmatrix}$$

Sol: จากโจทย์ จะได้เป็น $\begin{bmatrix} 2x+10 & 2y-6 \\ 10 & 14+4z \end{bmatrix} = \begin{bmatrix} 2 & 3-y \\ w+7 & 14+z \end{bmatrix}$

นั่นคือ
$2x+10 = 2 \rightarrow 2x = -8 \therefore x = -4$ #
$2y-6 = 3-y \rightarrow 3y = 9 \therefore y = 3$ #
$10 = w+7 \rightarrow 3 = w \therefore w = 3$ #
$14+4z = 14+z \rightarrow z = 0 \therefore z = 0$ ##

Ans.

2. Using elementary row operations to solve the linear equation

$$\begin{aligned} 2x_1 - x_2 &= 0, \\ -x_1 + 2x_2 + x_3 &= 6, \\ -x_1 - x_3 &= -4. \end{aligned}$$

Sol: จากโจทย์ จะได้เป็น

$$[A|B] = \left[\begin{array}{ccc|c} 2 & -1 & 0 & 0 \\ -1 & 2 & 1 & 6 \\ -1 & 0 & -1 & -4 \end{array}\right] \sim \overset{-R_3}{\left[\begin{array}{ccc|c} 2 & -1 & 0 & 0 \\ -1 & 2 & 1 & 6 \\ 1 & 0 & 1 & 4 \end{array}\right]} \overset{\substack{R_1-2R_3 \\ R_2+R_3}}{\sim} \left[\begin{array}{ccc|c} 0 & -1 & -2 & -8 \\ 0 & 2 & 2 & 10 \\ 1 & 0 & 1 & 4 \end{array}\right]$$

$$\overset{-R_1}{\sim} \left[\begin{array}{ccc|c} 0 & 1 & 2 & 8 \\ 0 & 1 & 1 & 5 \\ 1 & 0 & 1 & 4 \end{array}\right] \sim \overset{R_1-R_2}{\left[\begin{array}{ccc|c} 0 & 0 & 1 & 3 \\ 0 & 1 & 1 & 5 \\ 1 & 0 & 1 & 4 \end{array}\right]} \overset{\substack{R_2-R_1 \\ R_3-R_1}}{\sim} \left[\begin{array}{ccc|c} 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & 2 \\ 1 & 0 & 0 & 1 \end{array}\right]$$

$$\overset{R_3 \leftrightarrow R_1}{\sim} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array}\right] \Rightarrow \therefore x = 1, y = 2, z = 3 \#$$

Ans.

3. Find an <u>echelon form</u> of $Ax = b$ and solve the system:

$$\begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 6 & 1 \\ 3 & 9 & 2 \\ 0 & -1 & 3 \end{bmatrix} \text{ and } b = \begin{bmatrix} 5 \\ 8 \\ 3 \end{bmatrix}$$

Sol: ตามโจทย์ จะได้ว่า $[A|b] \to \left[\begin{array}{ccc|c} 2 & 6 & 1 & 5 \\ 3 & 9 & 2 & 8 \\ 0 & -1 & 3 & 3 \end{array}\right]$

หรือก็คือ $\left[\begin{array}{ccc|c} 2 & 6 & 1 & 5 \\ 3 & 9 & 2 & 8 \\ 0 & -1 & 3 & 3 \end{array}\right]$

$\sim R_2 - \frac{3}{2}R_1 \left[\begin{array}{ccc|c} 2 & 6 & 1 & 5 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & 3 & 3 \end{array}\right]$

$\sim \begin{matrix} R_1 - 2R_2 \\ R_3 - 6R_2 \end{matrix} \left[\begin{array}{ccc|c} 2 & 6 & 0 & 4 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & 3 & 3 \end{array}\right]$

$\sim R_2 \leftrightarrow R_3 \left[\begin{array}{ccc|c} 2 & 6 & 0 & 4 \\ 0 & -1 & 3 & 3 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{array}\right]$

จะได้ว่า $2x + 6y = 4$ แทนค่า $y=0$ $\longrightarrow 2x = 4 \longrightarrow x = 2$

$-y + 3z = 3$ แทนค่า $z=1$ $\longrightarrow -y + 3 = 3 \longrightarrow y = 0$

$\frac{1}{2}z = \frac{1}{2} \longrightarrow z = 1$

$\therefore x = 2, y = 0, z = 1$ // Ans.

4. Find an echelon form of $Ax = b$ and solve the system:

$$A = \begin{bmatrix} 0 & 2 & 5 \\ -1 & 7 & -5 \\ -1 & 8 & 3 \end{bmatrix} \text{ and } b = \begin{bmatrix} 15 \\ -20 \\ 4 \end{bmatrix}$$

Sol: ตามโจทย์ จะได้ว่า $[A|b] \to \left[\begin{array}{ccc|c} 0 & 2 & 5 & 15 \\ -1 & 7 & -5 & -20 \\ -1 & 8 & 3 & 4 \end{array}\right]$

เขียนก็คือ $\left[\begin{array}{ccc|c} 0 & 2 & 5 & 15 \\ -1 & 7 & -5 & -20 \\ -1 & 8 & 3 & 4 \end{array}\right]$

$\sim \begin{matrix} \\ R_3 - R_2 \end{matrix} \left[\begin{array}{ccc|c} 0 & 2 & 5 & 15 \\ -1 & 7 & -5 & -20 \\ 0 & 1 & 8 & 24 \end{array}\right]$

$\sim \begin{matrix} R_1 - 2R_3 \\ \\ \end{matrix} \left[\begin{array}{ccc|c} 0 & 0 & -11 & -33 \\ -1 & 7 & -5 & -20 \\ 0 & 1 & 8 & 24 \end{array}\right]$

$\sim -\frac{1}{11}R_1 \left[\begin{array}{ccc|c} 0 & 0 & 1 & 3 \\ -1 & 7 & -5 & -20 \\ 0 & 1 & 8 & 24 \end{array}\right]$

$\sim \begin{matrix} \\ R_3 - 8R_1 \end{matrix} \left[\begin{array}{ccc|c} 0 & 0 & 1 & 3 \\ -1 & 7 & -5 & -20 \\ 0 & 1 & 0 & 0 \end{array}\right]$

$\sim R_1 \leftrightarrow R_3 \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ -1 & 7 & -5 & -20 \\ 0 & 0 & 1 & 3 \end{array}\right]$

$\sim \begin{matrix} \\ R_3 - 7R_1 + 5R_3 \end{matrix} \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{array}\right]$

$\sim -R_2 \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right]$

$\sim R_1 \leftrightarrow R_2 \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{array}\right]$

$\therefore x = 5, y = 0, z = 3$ // Ans.

5. Find the sequences of elementary operations that will turn A to a reduce echelon matrix, where A is the following:

$$A = \begin{bmatrix} 3 & 0 & 2 & -1 \\ 2 & 3 & -1 & 0 \\ 1 & 5 & 0 & 8 \end{bmatrix}$$

Sol: จากโจทย์ จะได้ว่า

$$A = \begin{bmatrix} 3 & 0 & 2 & -1 \\ 2 & 3 & -1 & 0 \\ 1 & 5 & 0 & 8 \end{bmatrix} \xrightarrow{\frac{1}{3}R_1} \begin{bmatrix} 1 & 0 & \frac{2}{3} & -\frac{1}{3} \\ 2 & 3 & -1 & 0 \\ 1 & 5 & 0 & 8 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 0 & \frac{2}{3} & -\frac{1}{3} \\ 0 & 3 & -\frac{7}{3} & \frac{2}{3} \\ 0 & 5 & -\frac{2}{3} & \frac{25}{3} \end{bmatrix} \xrightarrow{\frac{1}{3}R_2} \begin{bmatrix} 1 & 0 & \frac{2}{3} & -\frac{1}{3} \\ 0 & 1 & -\frac{7}{9} & \frac{2}{9} \\ 0 & 5 & -\frac{2}{3} & \frac{25}{3} \end{bmatrix}$$

$$\xrightarrow{R_3 - 5R_1} \begin{bmatrix} 1 & 0 & \frac{2}{3} & -\frac{1}{3} \\ 0 & 1 & -\frac{7}{9} & \frac{2}{9} \\ 0 & 0 & \frac{29}{9} & \frac{65}{9} \end{bmatrix} \xrightarrow{\frac{9}{29}R_3} \begin{bmatrix} 1 & 0 & \frac{2}{3} & -\frac{1}{3} \\ 0 & 1 & -\frac{7}{9} & \frac{2}{9} \\ 0 & 0 & 1 & \frac{65}{29} \end{bmatrix} \xrightarrow[R_2 + \frac{7}{9}R_3]{R_1 - \frac{2}{3}R_3} \begin{bmatrix} 1 & 0 & 0 & -\frac{53}{29} \\ 0 & 1 & 0 & \frac{59}{29} \\ 0 & 0 & 1 & \frac{65}{29} \end{bmatrix}$$

$$\therefore A \sim \begin{bmatrix} 1 & 0 & 0 & -\frac{53}{29} \\ 0 & 1 & 0 & \frac{59}{29} \\ 0 & 0 & 1 & \frac{65}{29} \end{bmatrix}$$

// Ans.

6. Verify whether $Ax = b$ is consistent for the following A and b and find a solution if it exists:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -4 \\ 0 & 1 & -3 \\ -1 & 0 & -2 \end{bmatrix} \text{ and } b = \begin{bmatrix} 1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

Sol: จากโจทย์ จะได้ว่า $[A|b] \Rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 1 & 2 & -4 & -1 \\ 0 & 1 & -3 & -1 \\ -1 & 0 & -2 & -1 \end{array}\right]$

พิจารณา

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 1 & 2 & -4 & -1 \\ 0 & 1 & -3 & -1 \\ -1 & 0 & -2 & -1 \end{array}\right] \xrightarrow{-\frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 2 & -4 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 2 & -6 & -2 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 2 & -4 & -1 \\ 2 & 1 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ -1 & 0 & -2 & -1 \end{array}\right]$$

$$\xrightarrow[R_3 + R_2]{R_4 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & -4 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[R_4 + R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -4 & -1 \\ 0 & -3 & 9 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & 2 & -6 & -2 \end{array}\right]$$

ทำให้ได้ว่า $x_1 + 2x_2 - 4x_3 = -1$ —①

$x_2 - 3x_3 = -1$ —②

จาก ② ; $x_2 = 3x_3 - 1$

นำค่า x_2 แทนลงใน ① ; $x_1 + 6x_3 - 2 - 4x_3 = -1$

$x_1 = -2x_3 + 1$

ให้ $x_3 = t$

จะได้ $\vec{x} = t\begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

$\therefore \vec{x} = t\begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \ t \in \mathbb{R}$

7. Find the values of a for which the system

$$x_1 + x_2 - x_3 = 3$$
$$x_1 - x_2 + 3x_3 = 4$$
$$x_1 + x_2 + \left(a^2 - 10\right) x_3 = a$$

(1) has no solution (2) has the unique solution (3) has infinitely many solutions.

Sol!! จากโจทย์ จะได้ว่า $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 3 \\ 1 & -1 & 3 & 4 \\ 1 & 1 & a^2-10 & a \end{array}\right] \sim \begin{matrix} R_2-R_1 \\ R_3-R_1 \end{matrix} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 3 \\ 0 & -2 & 4 & 1 \\ 0 & 0 & a^2-9 & a-3 \end{array}\right]$

กรณี 1 : ไม่มีคำตอบ ถ้าระบบสมการไม่มีคำตอบ แสดงว่า

$a^2-9=0$ และ $a-3 \neq 0 \rightarrow a \neq 3$

นั่นคือ $(a-3)(a+3)=0 \longrightarrow a=-3, \cancel{3}$ $\therefore a=-3$ //

กรณี 2 : มีคำตอบเดียว ถ้าระบบสมการมีคำตอบเดียว แสดงว่า

$a^2-9 \neq 0$

นั่นคือ $(a-3)(a+3) \neq 0 \longrightarrow a \neq -3, 3$ $\therefore a \neq -3, 3$ //

กรณี 3 : มีหลายคำตอบ ถ้าระบบสมการมีหลายคำตอบ แสดงว่า

$a^2-9=0$ และ $a-3=0 \rightarrow a=3$

นั่นคือ $(a-3)(a+3)=0 \longrightarrow a=\cancel{-3}, 3$ $\therefore a=3$ // Ans.